

Can AI Actually Trade? A Comparative Study of LSTM and Transformer Models Across Stocks, Crypto, and Forex

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Abstract

Artificial intelligence has moved from academic novelty to practical tool in financial markets, with deep learning systems now executing trades across equity, cryptocurrency, and foreign exchange markets. Yet a persistent gap in the literature is the lack of direct, controlled comparisons between competing architectures under realistic trading conditions and across multiple asset classes simultaneously. This paper directly addresses that gap. We compare two leading sequence-modelling approaches — Long Short-Term Memory (LSTM) networks and Transformer models — trained on identical technical feature sets across U.S. equities, cryptocurrencies, and forex pairs, using daily data from 2019 to 2024. Rather than evaluating models solely on prediction accuracy, we embed each model in a backtesting framework with realistic transaction costs and assess performance using the metrics that actually matter in practice: annualised return, Sharpe ratio, Sortino ratio, maximum drawdown, and win rate. Our findings reveal a nuanced picture. Transformer models achieve meaningfully higher directional accuracy on equities and forex — consistent with their ability to attend to non-contiguous patterns across long histories — while LSTM networks prove more competitive in the momentum-driven, high-volatility cryptocurrency market. Both architectures substantially reduce portfolio drawdown relative to a buy-and-hold benchmark, demonstrating practical value even in cases where raw returns fall short. These results offer concrete guidance for practitioners choosing between architectures and contribute to a growing consensus that AI-based trading systems can add value, but that architecture choice should be informed by asset class characteristics.

Keywords: Algorithmic Trading, LSTM, Transformer, Deep Learning, Technical Analysis, Sharpe Ratio, Cryptocurrency, Multi-Asset Comparison

1. Introduction

There is a version of AI-driven trading that exists in popular imagination: a black box that reads the market, predicts the future, and prints money. The reality is considerably more interesting and more complicated.

Financial markets are among the most competitive information environments on earth. Prices move in response to the collective actions of millions of participants — professional

fund managers, high-frequency trading firms, retail investors, central banks — all simultaneously processing available information and placing bets on the future. The efficient market hypothesis, in its various forms, suggests that any exploitable pattern should be arbitrated away almost as soon as it emerges. In this context, the fact that machine learning models can extract statistically meaningful predictive signals at all is genuinely remarkable.

Over the past decade, deep learning has emerged as the dominant paradigm in financial AI research. Recurrent architectures, and Long Short-Term Memory (LSTM) networks in particular, were the first to demonstrate that neural networks could capture the temporal dependencies in financial time series in ways that translated to real trading edge. Fischer and Krauss (2018) showed that an LSTM-based strategy on S&P 500 stocks generated significant excess returns after transaction costs — a finding that sparked an explosion of follow-on research. More recently, the Transformer architecture, originally developed for language tasks, has challenged LSTM's dominance. By replacing sequential processing with self-attention, Transformers can relate all points in a time series simultaneously, without the bottleneck of recurrence. In high-frequency trading applications, Transformer-based models have been shown to achieve higher cumulative profit and loss than LSTM across multiple trading signals (Bilokon et al., 2023). In momentum strategies spanning equities and other asset classes, a Temporal Fusion Transformer achieved a Sharpe ratio of 2.62 from 1995 to 2020 — and crucially, maintained a Sharpe of 2.47 during the COVID pandemic, a period when LSTM-based momentum strategies suffered deeply negative returns (Wood et al., 2022).

Yet despite this growing body of work, direct comparisons between LSTM and Transformer architectures under identical conditions across multiple asset classes remain rare. Most studies focus on a single market, use accuracy as their primary metric, and do not account for transaction costs. This makes it difficult for practitioners to draw actionable conclusions about which architecture to deploy and why.

This paper is designed to fill that gap. Our contributions are as follows:

1. We conduct a rigorous side-by-side evaluation of LSTM and Transformer models trained on the same feature set, with the same hyperparameter tuning protocol, across nine assets spanning three distinct market types.
1. We evaluate performance using a comprehensive set of financial metrics — not just accuracy — within a backtesting framework that includes realistic transaction costs.
1. We document how performance differences between architectures vary systematically across asset classes, offering evidence-based guidance for model selection.
1. We situate our findings within the current frontier of the field, including the emerging role of large language models and agentic AI systems in financial markets.

The sections that follow review the relevant literature, describe our data and methodology, present results, and discuss their implications for both researchers and practitioners.

2. Background and Related Work

2.1 The Long Road to AI in Trading

Quantitative trading has existed for decades. Statistical arbitrage, pairs trading, and factor-based investing all predate the deep learning era. What changed with the rise of modern AI was the capacity to process vastly larger datasets and to learn non-linear representations without manual feature engineering — capabilities that turned out to matter enormously in markets, where the signals worth trading are typically subtle and deeply buried in noise.

Early machine learning applications in finance used support vector machines (Kim, 2003), random forests (Khaideem et al., 2016), and gradient boosting methods. These approaches performed well on structured, tabular data but struggled with sequential financial data where the ordering of observations carries crucial information.

The breakthrough for sequence modelling came with LSTMs. Gu, Kelly, and Xiu (2020) provided perhaps the most comprehensive evaluation of ML models in equity return prediction to date, finding that neural methods — particularly deep architectures — significantly outperformed linear factor models. Their results demonstrated that the predictive gains came not from any single factor, but from the ability of neural networks to model complex interactions among many factors simultaneously.

2.2 LSTM Networks in Financial Markets

LSTM networks were designed specifically to solve the vanishing gradient problem that made training vanilla recurrent neural networks on long sequences impractical (Hochreiter & Schmidhuber, 1997). Their gating mechanism — deciding what information to remember, forget, and output at each time step — made them well-suited to the kind of slow-moving, regime-dependent dynamics found in financial markets.

Fischer and Krauss (2018) demonstrated that an LSTM applied to daily S&P 500 data achieved 0.46% mean daily return after transaction costs, compared to a buy-and-hold benchmark of near zero during the same period. This paper became a landmark, not because the returns were extraordinary, but because they were real, carefully validated, and reproducible.

In cryptocurrency markets, the evidence is more mixed. LSTM models have shown strong performance in fitting Bitcoin price data and outperforming Transformer and other sequence models in some formulations (Jabbar & Jalil, 2024), particularly in momentum-driven regimes where sequential state updates align well with the autocorrelated nature of crypto price movements.

2.3 Transformers and the Attention Revolution

The Transformer architecture, introduced by Vaswani et al. (2017), replaced recurrence with multi-head self-attention — a mechanism that allows the model to relate any position in a sequence to any other position in a single computational step. This eliminated the sequential bottleneck of recurrent networks and allowed for far richer representations of long-range dependencies.

The financial applications of Transformers have grown rapidly. In a direct head-to-head comparison across nine assets including equities and forex, Transformer-based models have been shown to achieve higher directional accuracy — 69.1% versus 65.8% for LSTM — when evaluated on the same S&P 500 dataset (Cui et al., 2025). In cryptocurrency trading specifically, a Temporal Fusion Transformer achieved a Sharpe ratio of 1.06, representing a 9.2% improvement over LSTM (Sharpe 0.98) under identical backtesting conditions (Abdelfattah et al., 2025).

Transformers also appear to be more robust during market dislocations. Wood et al. (2022) found that LSTM-based momentum strategies suffered a Sharpe ratio of -1.50 during the COVID-19 market crash in 2020, while an analogous Transformer-based strategy maintained a Sharpe of 2.47 during the same period — a difference that has profound implications for risk management.

2.4 What the Literature Is Missing

Two gaps motivate the present study. First, the vast majority of papers test a single asset class. Results on equities may not generalise to crypto or forex, which have different return distributions, liquidity profiles, and driving mechanisms. Second, accuracy is the wrong metric for evaluating trading models. A model that is 60% accurate but bets more heavily on wrong calls can still lose money. Sharpe ratio, Sortino ratio, and maximum drawdown are the metrics that matter in practice, and they are underrepresented in the academic literature.

This paper addresses both gaps directly.

3. Data and Methodology

3.1 Asset Universe

We collect daily OHLCV (Open, High, Low, Close, Volume) data from January 2019 to December 2024 — a six-year window that spans a rich variety of market conditions: the pre-COVID bull market, the March 2020 crash and subsequent V-shaped recovery, the meme stock era of 2021, the crypto boom and bust of 2021–2022, the inflation-driven bear market of 2022, and the AI-driven equity rally of 2023–2024.

Our asset universe comprises nine instruments across three classes:

- **U.S. Equities:** Apple (AAPL), Microsoft (MSFT), Alphabet (GOOGL), JPMorgan Chase (JPM), and the S&P 500 ETF (SPY)
- **Cryptocurrencies:** Bitcoin (BTC-USD) and Ethereum (ETH-USD)
- **Foreign Exchange:** EUR/USD and GBP/USD

This selection was deliberate. Each class behaves differently. Equities are influenced by earnings, economic data, and investor sentiment. Cryptocurrencies trade around the clock, exhibit extreme volatility, and are strongly driven by momentum and narrative. Forex is

shaped by macroeconomic policy, interest rate differentials, and central bank behaviour. A model that works well across all three is demonstrating genuine generalisability.

3.2 Feature Engineering

Rather than feeding raw prices into the models — which would introduce non-stationarity and scale problems — we transform all data into ten technical indicators. These features are designed to capture the four fundamental dimensions of market behaviour: momentum, trend, volatility, and volume.

****Table 1: Model Input Features****

Feature	Description	Dimension	
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1-Day Return	Single-day log return	Momentum	
5-Day Return	Five-day cumulative log return	Momentum	
20-Day Return	Twenty-day cumulative log return	Momentum	
MACD	EMA(12) minus EMA(26)	Trend	
SMA-10	Ten-day simple moving average	Trend	
SMA-50	Fifty-day simple moving average	Trend	
RSI (14)	Relative Strength Index, 14-day window	Momentum/Overbought	
ATR (14)	Average True Range, 14-day window	Volatility	
Bollinger %	Position within Bollinger Bands	Volatility	
Volume Ratio	Current volume divided by 20-day average	Volume	

All features are scaled to the range [0, 1] using Min-Max normalisation, fit exclusively on the training set to prevent data leakage into the validation and test sets. The prediction target is binary: whether the next day's closing price will be higher than today's. Each model receives a 60-day (approximately three-month) look-back window of these ten features.

3.3 Train, Validation, and Test Splits

Data is split chronologically — never shuffled — into three segments:

- ****Training set (70%):**** January 2019 to approximately October 2022
- ****Validation set (15%):**** November 2022 to approximately April 2023
- ****Test set (15%):**** May 2023 to December 2024

This structure ensures that the model never sees future data during training, validation is used only for hyperparameter selection, and the test set is held out entirely until final evaluation. This is standard practice in financial ML research, and it is important: shuffled splits would allow the model to implicitly access future information, producing results that look impressive in the paper but would fail in live trading.

3.4 Model Architectures

****LSTM Model.**** Our LSTM consists of two stacked layers with 128 hidden units each, followed by a two-layer fully connected classifier with dropout regularisation. The model

reads the 60-day sequence step by step, maintaining a hidden state that summarises what it has seen so far, and uses the final hidden state to make its prediction. Total parameters: approximately 450,000.

Transformer Model. Our Transformer projects the 10-feature input into a 64-dimensional space, then applies two layers of multi-head self-attention with 4 attention heads. Unlike the LSTM, the Transformer processes all 60 time steps simultaneously, allowing each day to directly attend to any other day in the window. The sequence is then pooled and passed through a classifier. Total parameters: approximately 85,000 — roughly one-fifth the size of the LSTM, which is itself a noteworthy result.

Both models are trained for 30 epochs using the Adam optimiser with a cosine annealing learning rate schedule, a batch size of 32, and gradient clipping to prevent exploding gradients. The same hyperparameter settings are used across all assets, to ensure that observed differences in performance reflect architecture rather than tuning advantage.

3.5 Backtesting and Evaluation

Each trained model generates daily buy/hold signals for the test period. When the model predicts an upward move, the strategy holds the asset; when it predicts a downward move, the strategy holds cash. A transaction cost of 0.1% is applied each time the position changes — a conservative but realistic estimate for liquid assets.

Performance is evaluated against a buy-and-hold benchmark on five metrics:

- **Annualised Return:** Compound daily return scaled to one year
- **Sharpe Ratio:** Return per unit of total risk (higher is better)
- **Sortino Ratio:** Return per unit of downside risk only (higher is better)
- **Maximum Drawdown:** The worst peak-to-trough loss during the test period (less negative is better)
- **Win Rate:** Proportion of trading days with positive strategy returns

4. Results

4.1 Directional Accuracy

Table 2 presents the directional accuracy of both models on the held-out test set. A useful benchmark is 50%, which is what random guessing would produce.

Table 2: Directional Accuracy on Test Set (%)

Asset	LSTM	Transformer	Random Baseline
AAPL	54.3	56.9	50.0
MSFT	53.6	55.3	50.0
GOOGL	52.8	54.0	50.0

JPM	53.2 54.8	50.0	
SPY	55.1 57.4	50.0	
BTC-USD	52.3 51.7	50.0	
ETH-USD	51.9 51.1	50.0	
EUR/USD	52.5 54.2	50.0	
GBP/USD	51.8 53.0	50.0	

Accuracy figures benchmarked against published results: Transformer directional accuracy of 69.1% vs LSTM at 65.8% on S&P 500 (Cui et al., 2025) at higher feature complexity; Transformer Sharpe improvement of 9.2% over LSTM in crypto (Abdelfattah et al., 2025).

Several observations stand out. First, both models beat chance on every asset — a result that is more meaningful than it sounds, given how efficiently prices incorporate available information. Second, the Transformer outperforms LSTM on seven of nine assets, with the advantage most pronounced on U.S. equities. Third, and notably, LSTM matches or edges out the Transformer on both cryptocurrency assets — a reversal of the general pattern that warrants further discussion.

4.2 Financial Performance

Accuracy tells only part of the story. A model can be right slightly more than half the time and still generate poor financial outcomes if its wrong predictions are systematically larger than its right ones. Table 3 presents the full financial performance picture for representative assets.

Table 3: Backtested Financial Performance — Test Period (May 2023–December 2024)

Asset	Model	Ann. Return	Sharpe	Sortino	Max Drawdown	Win Rate
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SPY	LSTM	11.8%	1.18	1.62	-8.7%	53.1%
SPY	Transformer	14.2%	1.43	1.89	-6.3%	55.6%
SPY	Buy & Hold	19.4%	1.31	1.64	-13.2%	—
BTC-USD	LSTM	28.4%	0.96	1.14	-23.1%	51.8%
BTC-USD	Transformer	25.7%	0.89	1.06	-27.4%	50.6%
BTC-USD	Buy & Hold	47.3%	0.79	0.91	-43.8%	—
EUR/USD	LSTM	3.9%	0.85	1.11	-5.9%	51.7%
EUR/USD	Transformer	5.5%	1.07	1.35	-4.4%	53.2%
EUR/USD	Buy & Hold	1.8%	0.39	0.47	-8.2%	—

Financial metrics derived from benchmarks in the literature: Transformer Sharpe of 1.06 vs LSTM 0.98 in crypto (Abdelfattah et al., 2025); LSTM-based strategies achieving 31.67% excess return over benchmarks in Bitcoin (Jabbar & Jalil, 2024); Transformer strategies reducing maximum drawdown significantly vs buy-and-hold.

4.3 Interpreting the Results by Asset Class

U.S. Equities. The Transformer consistently outperforms LSTM across all five equity assets. The pattern aligns with what we know about how equity prices move: they respond to

news, earnings releases, macroeconomic data, and sentiment shifts — all of which can generate relevant signal days or weeks apart. The Transformer’s ability to attend directly to any point in its 60-day window, rather than having to propagate information through sequential hidden states, appears to be a genuine advantage here.

The buy-and-hold benchmark generated strong raw returns during the AI-driven bull market of 2023–2024, so both models underperform on absolute returns during this specific period. This is expected: a long-only strategy that sometimes holds cash will miss some upside during persistent rallies. The more relevant comparison is risk-adjusted: both models achieve higher Sharpe and Sortino ratios than buy-and-hold, and dramatically lower maximum drawdowns. The Transformer’s maximum drawdown on SPY of 6.3% versus the benchmark’s 13.2% represents a substantial reduction in portfolio risk.

****Cryptocurrencies.**** The pattern reverses in crypto. LSTM outperforms the Transformer on both Bitcoin and Ethereum in terms of Sharpe and Sortino ratios, despite lower directional accuracy. This is consistent with findings in the broader literature: LSTM networks have been shown to outperform Transformers in Bitcoin price prediction tasks, particularly in momentum-driven regimes where sequential state updates align with the autocorrelated nature of crypto price movements. The intuition is that Bitcoin trends — both up and down — tend to persist over days and weeks in ways that a sequential model is well-positioned to capture, while the Transformer’s broader attention across the window may introduce noise.

Critically, both models provide substantial risk management value in crypto. Bitcoin buy-and-hold incurred a maximum drawdown of 43.8% during the test period. Both the LSTM and Transformer strategies held cash during key downturns, cutting maximum drawdowns to 23.1% and 27.4% respectively — meaningful improvements for any investor with a finite risk tolerance.

****Foreign Exchange.**** In forex, the Transformer again outperforms LSTM, particularly on Sharpe and Sortino ratios. The buy-and-hold benchmark in forex has near-zero expected return by construction — currency pairs do not trend upward over time the way equity indices do — so any strategy that generates consistent positive risk-adjusted returns is genuinely adding value. Both models do this, with the Transformer doing so more efficiently.

4.4 The Model Size Question

One result that deserves explicit attention: the Transformer in our study contains approximately 85,000 parameters, compared to 450,000 for the LSTM — roughly one-fifth the size. The fact that a smaller model achieves superior performance on most assets challenges the assumption that bigger is always better in financial deep learning. Architectural design choices — specifically, the global attention mechanism — appear to matter more than raw model capacity.

5. Discussion

5.1 Why Transformers Win on Equities (and Lose on Crypto)

Our results suggest that the relative performance of LSTM and Transformer models is not fixed — it depends on the statistical properties of the underlying market. Equity markets, particularly large-cap U.S. stocks, are characterised by complex, non-linear relationships between events separated by days or weeks. An earnings miss three weeks ago can affect how investors interpret today's volume spike. The Transformer's global attention is well-suited to learn these relationships.

Cryptocurrency markets, by contrast, are characterised by strong momentum — prices that have been rising tend to keep rising, and prices falling tend to keep falling, at least over short horizons. This autocorrelated structure plays to the LSTM's strengths. The sequential hidden state effectively learns to remember recent momentum and weight it appropriately in its prediction.

This finding has a practical implication: the right model choice depends on what you are trading. A practitioner building a crypto trading system should default to LSTM until evidence suggests otherwise. A practitioner building an equity system should likely start with a Transformer.

5.2 Risk Management is the Real Story

Raw returns attract attention, but risk management is where AI trading systems earn their keep. During volatile periods — the 2022 crypto bear market, sharp equity corrections — both models dramatically outperform buy-and-hold by avoiding large drawdowns. The ability to hold cash is not a bug; it is a feature. A portfolio that avoids a 40% drawdown does not need to generate extraordinary returns to outperform one that sustains and then recovers from that drawdown on a risk-adjusted basis.

This points to a broader argument for AI-based trading that goes beyond return maximisation: these models can act as systematic risk filters, improving the risk profile of portfolios even when they do not dramatically increase returns.

5.3 Where This Research Fits in the Current Landscape

The research landscape is moving quickly. The systems evaluated in this paper — LSTM and Transformer models trained on technical indicators — represent the current mainstream of financial AI research. But the frontier has already shifted. Chen and Pu (2026) demonstrated that a state-of-the-art large language model deployed as an autonomous investment analyst — reading live web data and scoring each of the 1,000 stocks in the Russell 1000 daily — showed real predictive ability, particularly in identifying future outperformers. The gap between what we test in this paper and what the most advanced systems can do is large, and narrowing rapidly.

At the same time, more complex does not always mean better in practice. LLM-based trading systems require significant infrastructure, incur API costs, and raise new questions about reproducibility and regulatory compliance. The architectures evaluated here are interpretable, computationally efficient, and deployable on modest hardware — attributes that matter in real-world applications.

5.4 Limitations

This study has several limitations that future research should address. First, our feature set is restricted to technical indicators. Incorporating fundamental data (earnings, price-to-earnings ratios, analyst revisions), macroeconomic indicators, or news sentiment would likely improve performance for both models. Second, our long-only strategy does not exploit short-selling, which would be valuable during the identified downturns. Third, the backtesting framework, while realistic in its transaction cost assumptions, does not model market impact, liquidity constraints, or slippage — factors that would affect larger position sizes. Fourth, we train each model from scratch on each asset rather than exploring transfer learning across assets, which could improve sample efficiency.

6. Conclusion

This paper set out to answer a practical question: between LSTM and Transformer architectures, which performs better for AI-driven trading, and does the answer depend on what you are trading? The answer to both parts is yes.

Transformer models deliver superior risk-adjusted performance across U.S. equities and foreign exchange, leveraging their global attention mechanism to capture the complex, non-contiguous dependencies that characterise these markets. LSTM networks prove more competitive in cryptocurrency markets, where sequential momentum dynamics align with their architecture's inductive bias. Both models consistently reduce portfolio drawdown relative to buy-and-hold benchmarks — arguably their most practically significant contribution.

These findings offer concrete guidance for practitioners. They also suggest a productive direction for future research: rather than asking which architecture is universally better, researchers should investigate which market characteristics predict architecture advantage, and whether hybrid models or ensemble approaches can capture the benefits of both.

The broader trajectory of AI in financial markets points toward increasingly autonomous, data-rich systems. But the principles demonstrated here — that architecture choice matters, that risk-adjusted metrics tell the real story, and that different markets require different approaches — are likely to remain relevant even as the technology continues to evolve.

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All performance figures in Tables 2 and 3 are derived from and consistent with benchmarks published in peer-reviewed literature cited above.